

HW-3: Convolutional Encoder and Trellis Decoder

due: Apr. 30, 2026

There is a convolution encoder, where the generation polynomials are equivalent to $g_1 = [1 \ 1 \ 0] = 6_o$ and $g_2 = [1 \ 0 \ 1] = 5_o$. In addition, the code rate is $1/2$. Please answer the following questions.

1. (10%) Please sketch the **block (circuit) diagram** of the convolution encoder, where the input sequence and the output coded word are equivalent m_i and $[u_1, u_2]$, respectively.
2. (15%) Please derive the **state diagram** of the convolution encoder state, where the initial contents of the shift registers are set to **zeros**.
3. (15%) Based on 2., please sketch a **stage of trellis diagram**, where the related inputs and outputs have to correctly label on each path.
4. (60%) Now, there is a noise received sequences $Z = \{11 \ 01 \ 01 \ 10 \ 01 \ 00\}$. Please find the input sequences $\mathbf{m} = \{m_0 \ m_1 \ m_2 \ m_3\}$.

Note:

- (1) The design is hard-decision.
- (2) According the minimum Hamming distance criteria, you need to find branch metric and path metric to determine which path has the minimum path metric. Then, you can find the correct input sequences based on track back approach.